MIPS FloorPlan Assignment

FloorPlan script:

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#######################################################################
#       ASIC PHYSICAL DESIGN AUTOMATION SCRIPT            #
#######################################################################
# Author         : Khaled Gamal
# Project        : MIPS16
# Block/Top      : mips_16
# Technology     : saed90nm
# EDA Tool       : Synopsys (ICC2)
# Usage          : icc2_shell -f script/floorplan.tcl -output_log
log/floorplan.log
# Prerequisites : Design Library created
# Creation Date : 2026-08-20
# Educational    : ITI_summer_camp_2026_R3
#######################################################################


#######################################################################
    # --------------- Variable setup ------------- #
#######################################################################

set design  "mips_16"
set std_worst   "saed90nm_max.db"
set target_library $std_worst
set corner  "worst"

set project_dir "/mnt/hgfs/Shared_files/${design}"
set Lib_path    "/home/ICer/Downloads/Lib"
set dlib_path   "${project_dir}/pnr/2_design_lib/library"

set prev_stage      "dlib"
set current_stage   "floorplan"

#######################################################################
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    # --------------- dlib setup ------------- #
######################################################################

open_block  ${dlib_path}/${design}.dlib:${design}_${prev_stage}.design

# copying the (clean) block from the previous stage (dlib) to just
preserve it and work on the new one which is the present stage (floorplan)
#to make sure if any errors happened through the current stage, it will be
easy to restore the previous stage
copy_block -from_block ${design}.dlib:${design}_${prev_stage}.design
-to_block ${design}_${current_stage}

current_block ${design}_${current_stage}.design
start_gui

######################################################################
    # --------------- Layers setup ------------- #
######################################################################
# Metal layers directions
set_attribute [get_layer {M1 M3 M5 M7 M9}] routing_direction horizontal
set_attribute [get_layer {M2 M4 M6 M8 MRDL}] routing_direction vertical;
# MRDL (Metal Redistribution Layer) is an extra-thick layer on the
absolute top of the die used for power supply

# site def attribute (get unit tile name from the technology file)
set Name_unit [get_site_defs]
set_attribute $Name_unit is_default true

# flipping the unit tiles on the Y-axis to make common Vdd and Vss between
every two site rows
set_attribute $Name_unit symmetry {Y}

# set_attribute [get_layers {M1}] track_offset 0.037

######################################################################
    # --------------- Initialize Floorplan ------------- #
######################################################################

# All parameters related to core and die
initialize_floorplan    -control_type core \
```

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                        -core_utilization 0.6 \
                        -shape R \
                        -core_offset {10} \
                        -flip_first_row true \
                        -side_ratio {1 1}

#######################################################################
    # --------------- Placement Pins ------------- #
#######################################################################
# Auto pins placement (not preffered, manual is better)
place_pins -self -ports [get_ports *]

#######################################################################
    # --------------- Blockage Placement ------------- #
#######################################################################
#create_placement_blockage -boundary {{30 30} {50 50}} -name B1 -type hard

#create_placement_blockage -boundary {{5 3} {7 5}} -name B2 -type partial
-blockage_percentage 40

#create_placement_blockage -boundary {{7 3} {9 5}} -name B3 -type soft

#######################################################################
    # --------------- TAPCell Placement ------------- #
#######################################################################
#create_tap_cells -lib_cell [get_lib_cell */SAEDVT14_TAPDS] -pattern
stagger -distance 40

#seizeof_collection [get_cells tap*]
# remove_cell tap*

#create_tap_cells -lib_cell [get_lib_cell */SAEDVT14_TAPDS] -pattern
every_row -distance 10
#get_cells tap*
# remove_cell tap*

#create_tap_cells -lib_cell [get_lib_cell */SAEDVT14_TAPDS] -pattern
every_other_row -distance 10
#size_of[get_cells tap*]
# remove_cell tap*
```

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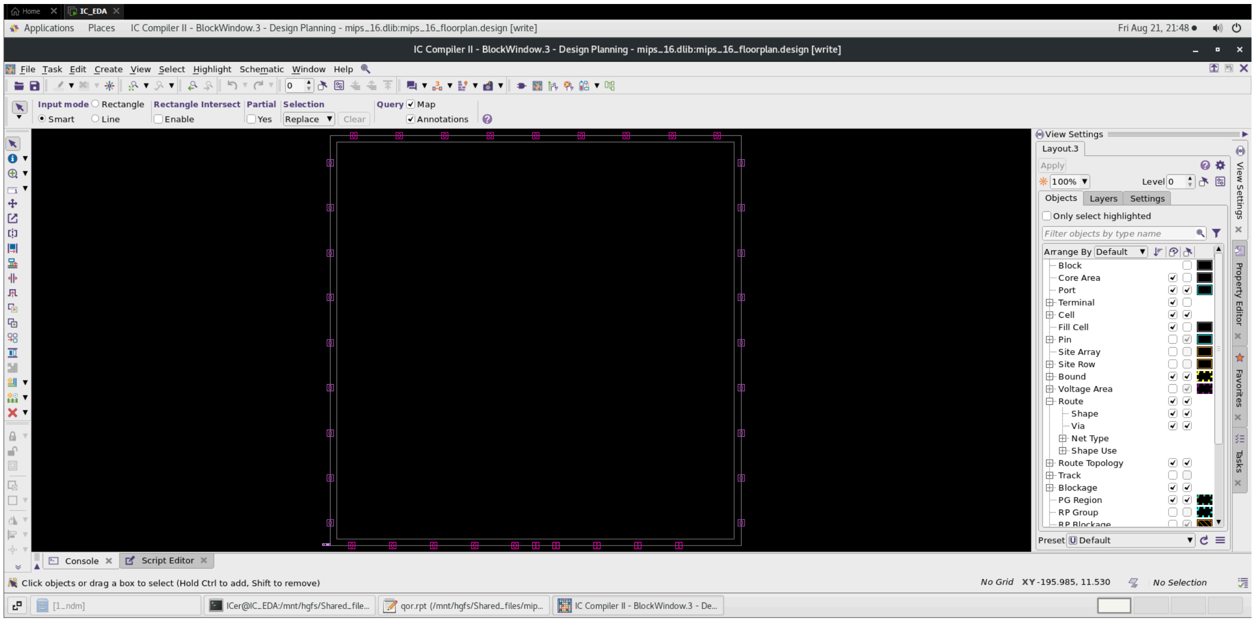
#######################################################################
    # --------------- Files Handling ------------- #
#######################################################################
sh rm    -r ../results/reports
sh rm    -r ../results/outputs
sh mkdir -p ../results/reports
sh mkdir -p ../results/outputs


#######################################################################
    # --------------- Reports ------------- #
#######################################################################
report_qor                  > ../results/reports/qor.rpt
report_utilization          > ../results/reports/utilization_default.rpt
get_placement_blockages     > ../results/reports/blockage.rpt


#######################################################################
    # --------------- Save Block ------------- #
#######################################################################
save_block -as ${design}_${current_stage}
${design}.dlib:${design}_${current_stage}.design
```

GUI screenshot:



Reports:

- <u>Blockage report</u>

Warning: No placement_blockage objects matched '*' (SEL-004)
Error: Nothing matched for collection (SEL-005)

- <u>qor report:</u>

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Report : qor
Design : mips_16
Version: O-2018.06-SP1
Date : Fri Aug 21 21:31:55 2026

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Timer Settings:
Delay Calculation Style: auto
Signal Integrity Analysis: disabled
Timing Window Analysis: disabled
Advanced Waveform Propagation: disabled
Variation Type: fixed_derate
Clock Reconvergence Pessimism Removal: disabled

Advanced Receiver Model: disabled
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Information: Corner default: no PVT mismatches. (PVT-032)
Information: The stitching and editing of coupling caps is turned OFF for design 'mips_16.dlib:mips_16_floorplan.design'. (TIM-125)
Information: Design Average RC for design mips_16_floorplan (NEX-011)
Information: r = 0.515372 ohm/um, via_r = 0.858630 ohm/cut, c = 0.062769 ff/um, cc = 0.000000 ff/um (X dir) (NEX-017)
Information: r = 0.562500 ohm/um, via_r = 0.870988 ohm/cut, c = 0.072228 ff/um, cc = 0.000000 ff/um (Y dir) (NEX-017)
Information: The RC mode used is VR for design 'mips_16'. (NEX-022)
Information: Update timing completed net estimation for all the timing graph nets (TIM-111)
Information: Net estimation statistics: timing graph nets = 13879, routed nets = 0, across physical hierarchy nets = 0, parasitics cached nets = 0, delay annotated nets = 0, parasitics annotated nets = 0, multi-voltage nets = 0. (TIM-112)

Scenario 'default'
Timing Path Group 'fun_clk'

Levels of Logic: 3
Critical Path Length: 4.90
Critical Path Slack: 8.60
Critical Path Clk Period: 15.00
Total Negative Slack: 0.00
No. of Violating Paths: 0
Worst Hold Violation: 0.00
Total Hold Violation: 0.00
No. of Hold Violations: 0

Cell Count

Hierarchical Cell Count: 11
Hierarchical Port Count: 335
Leaf Cell Count: 13831
Buf/Inv Cell Count: 1974
Buf Cell Count: 919
Inv Cell Count: 1055
CT Buf/Inv Cell Count: 0
Combinational Cell Count: 9591
Sequential Cell Count: 4240

Macro Count: 0

Area

Combinational Area: 103505.82
Noncombinational Area: 106566.45
Buf/Inv Area: 15067.24
Total Buffer Area: 8672.26
Total Inverter Area: 6394.98
Macro/Black Box Area: 0.00
Net Area: 0
Net XLength: 0.00
Net YLength: 0.00

Cell Area (netlist): 210072.27
Cell Area (netlist and physical only): 210072.27
Net Length: 0.00

Design Rules

Total Number of Nets: 13903
Nets with Violations: 0
Max Trans Violations: 0
Max Cap Violations: 0

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- **Utilization report:**

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Report : report_utilization
Design : mips_16
Version: O-2018.06-SP1
Date : Fri Aug 21 21:31:57 2026
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Utilization Ratio: 0.6014
Utilization options:
- Area calculation based on: site_row of block mips_16_floorplan

- Categories of objects excluded: hard_macros macro_keepouts soft_macros io_cells hard_blockages

Total Area:	349327.8720
Total Capacity Area:	349327.8720
Total Area of cells:	210072.2688
Area of excluded objects:	
- hard_macros :	0.0000
- macro_keepouts :	0.0000
- soft_macros :	0.0000
- io_cells :	0.0000
- hard_blockages :	0.0000

Utilization of site-rows with:
- Site 'unit': 0.6014

0.6014